

Andre Arante

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EDUCATION

The George Washington University

Washington, DC

Bachelor of Science in Electrical and Computer Engineering

Aug 2023 – May 2027

- Minor in Computer Science; coursework in Computer Vision, Reinforcement Learning, and Machine Learning

PUBLICATIONS

1. **A. Arante**, M. Ghazanfari, C. Vu, and P. Wei, “Open-Vocabulary Obstruction Detection for UAS Landing via Fine-Tuned YOLO-World” submitted to *AIAA SciTech 2027*, under review.

WORK EXPERIENCE

Software Engineering Intern

May 2026 – Present

Dell Technologies

Austin, TX

- Fine-tuned vision-language-action foundational models (π_0 and SmolVLA) via LoRA on curated demonstration datasets for contact-rich manipulation tasks, building the data collection, curation, and SFT pipeline end-to-end in order to achieve autonomous hardware validation with robotic arms.
- Extended post training with human-in-the-loop RL, training a residual policy that learns corrective actions from real-time human feedback, demonstrating improved task success where supervised fine-tuning alone fails

Undergraduate Research Intern

Sep 2025 – Present

The George Washington University

Washington, DC

- Ran experiments on the tradeoff between domain fine-tuning and open-vocabulary retention for open-vocabulary object detection models such as Yolo World and deployed for real-time, safety-critical landing zone obstruction detection for package delivery UAS drones in urban environments, achieved +149% mAP over baseline.

Software Engineering Intern

Feb 2024 – Feb 2025

C-3 Communication Systems, LLC

Bethesda, MD

- Owned end-to-end development of an autonomous UGV platform on Linux, architecting a modular software stack that integrated computer vision, obstacle detection, and autonomous navigation into a unified control system.
- Shipped and maintained a full-stack Operations Management System (HTML/CSS/JS, Firebase) in production for 1+ year, automating payroll, time tracking, and logistics workflows for a small startup

Data Science Intern

Jun 2023 – Aug 2023

Kaiser Permanente

San Jose, CA

- Refactored a SAS data pipeline into an automated daily job processing millions of patient records across 5 hospitals and generated informative Tableau dashboards, eliminating a manual workflow that gated critical staffing decisions

PROJECTS

SortBots – Distributed Multi-Robot Package Logistics (Senior Capstone)

- One of only 3 ECE student-proposed capstones selected from 19 submissions; only capstone project to independently secure an external sponsor to fully fund my project idea
- Designing a decentralized multi-robot coordination system for autonomous package logistics in warehouse environments using decentralized coordination, 3D perception (VSLAM), and multi-agent path planning

ImitationArm - Custom 6-DOF Robotic Arm Manipulator

- Developed a ROS 2 / C++ control stack with forward/inverse kinematics and MoveIt2 for motion planning on a high-precision 6-DOF manipulator; integrated MuJoCo for sim-to-real RL training
- Built a custom wearable IMU-based teleoperation controller (ESP32 + MPU-6050) for real-time teleoperation

Reinforcement Learning Agents in Minecraft

- Trained agents via Behavioral Cloning from human demonstrations using custom Python scripts to record Minecraft gameplay, then fine-tuned with actor-critic networks (PPO) to perform Minecraft parkour and bridging tasks

TECHNICAL SKILLS

ML & Foundation Models: PyTorch, CUDA, fine-tuning (SFT, RL, LORA), Weights & Biases, vision-language models (CLIP, YOLO-World), vision-language-action models (π_0 , SmolVLA)

Robotics & Deployment: C/C++, Python, Rust, ROS 2, MuJoCo, NVIDIA Jetson, Linux, Git, Docker, Kinematics